

# Towards Utilizing Domain Expertise for Exploratory Data Analysis

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## ABSTRACT

In exploratory data analysis, domain knowledge and experience play a central role in order to extract information from the data and to derive proof and knowledge. However, experienced domain experts are rarely the same people who carry out the data analyses. Therefore, utilizing domain expertise for guidance in analytic processes is a complex challenge. In recent years, machine learning has seen great advances. Increasing processing power and growth in data as well as affordable storage have led to more advanced algorithms. Therefore, with the emergence of applicable machine learning algorithms, there is now a method for preserving and making use even of complex knowledge. In this paper, we present a concept that allows to extract and utilize domain knowledge for exploratory data analysis. We introduce concepts of interaction store and analysis context store to record user interaction and context during an exploratory analysis. We use the recorded data to construct semantic interaction sequences and predict their potential insight. The prediction can then be used to guide other data scientist in their sensemaking while performing exploratory data analysis in similar domains and use cases. Furthermore, we discuss possible research opportunities and implications resulting from the presented concept.

## CCS CONCEPTS

• **Human-centered computing** → **Visual analytics**; • **Mathematics of computing** → *Exploratory data analysis*.

## KEYWORDS

User Interaction, Visual Analytics, Analytic Provenance, Exploratory Data Analysis, Machine Learning

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## 1 INTRODUCTION

One of the most common tasks of a data scientist is to find interesting insights and to discover knowledge in data sets by performing exploratory data analysis (EDA). During this process, the scientist examines the data to find patterns, detect anomalies (e.g. outliers) and test hypotheses. This is often done by using visual analytics (VA) - a method that combines statistical methods and interactive visualization tools to generate insight from data sets [14]. In this type of data analysis, the quality of findings heavily depends on the analyst's ability to ask the right questions and to perform a sequence of analysis interactions to make sense of the data and answer these questions. This sensemaking process is very complex and based on a synergy of human cognition, domain expertise and experience. Getting behind how sensemaking works has been part of research for several years and scientists have developed models to describe the process [13]. Nevertheless, much research is done in the area of how analysts generate insight [2]. User interaction seems to play an important role, since a lot of the analysts' mental models and cognitive processes are reflected in the interactions performed during the sensemaking process [15]. Recently, the research of tracking and understanding analysis interactions has become an increasingly important research topic known as *analytic provenance* [20]. Analytic provenance tries to answer questions from data visualization, capturing and encoding interactions in a meaningful way, recovering sensemaking processes from interactions through to transferability of recovered learnings into new domains. While many approaches of analytic provenance are concerned with tracking and recording user interactions [17, 22], there are other publications dealing with the analysis of interaction paths to understand the cognitive processes of analysts that lead to interesting insights [4, 5, 18], or to use user interactions for interactive model steering and active learning [3, 24].

Due to the complexity of the task to record and analyze a sensemaking process, there are not many approaches yet that try to transfer generated insights into guidance for analysis use cases [23]. Therefore, in this paper we want to present a concept for an approach to utilize recorded user interactions during EDA. We record experts' sensemaking processes by storing the interaction sequences and analysis context, and allow them to transfer findings into a final report to indicate insights. Based on the collected data, we generate candidate interaction sequences and rank them by predicting the insight from performing the sequence. The sequences with the highest predicted insight are then used for visual guidance of similar analyses. We will contribute a basic idea and system design for further research in this area and then discuss requirements, opportunities and challenges of such a system.

## 2 RELATED WORK

The presented concept builds on previous work in different areas. We will give an overview of the state of the art and related work of commercial visual VA systems, work in the area of understanding the role of humans in data analytics as well as previous approaches in analytic provenance and integration of user interaction with machine learning.

In 2018, Behrisch et al. [1] conducted a study on the status of commercial VA systems for the new requirements in the area of Big Data Analysis and concluded that most systems focus on confirmatory data analysis, where an analyst pursues a question and tries to find answers to this question within data. This traditionally happens in business analysis within a company, e.g. if the composition of a company's profit has to be reconstructed and reported. Behrisch et al. then argue that at present, however, VA systems are not particularly good at supporting analysts in exploratory analysis, especially in the area of Big Data, where it is necessary to search for the most important insights in all available data. There are few suitable guidance approaches that lead an analyst to the most important insights with the biggest impact on business.

In order to build a VA system it is important to understand how humans interact with such a system, as much can be learned about the mental model and the cognitive process [15]. Based on the results, it is then possible to create interaction possibilities that allow a person to interact better with a system. There is some work on the sensemaking process itself [27] and work on understanding and characterizing user interactions [12]. Furthermore, Ceneda et al. [7] formalize automated guidance in VA that defines the knowledge gaps as well as inputs and outputs to derive a guidance degree. North [19] presents an approach on how to measure insight from visualizations. From this area of related work, we can see approaches for formalizing user interaction and insight.

In the area of analytic provenance, there are a number of papers that deal with recording and analyzing user interaction e.g. [17, 22]. Endert et al. [9] introduce a system by defining interactions for similarity analysis between texts and assigning semantics to these interactions. The result is an expert system that can steer the underlying model on the basis of these semantic interactions. Feng et al. [10] and Gotz et al. [11] record user interactions and then utilize the interaction history in VA to integrate indicators into an analytics tool. This allows users to track their sensemaking process and revisit central views. Camisetty et al. [5] present similar work that uses analytic provenance data to enhance a web-based analytics application to record user interactions, lets them recap their interaction paths and allows them to revisit important choices when new data becomes available. There are additional related works that record user interaction and try to prepare and visualize user interactions in such a way that it becomes recognizable where an analyst has gained central insights [4, 18]. From the related work we can deduce methods to record user interactions, interaction semantics and approaches for quantifying insight.

To improve initial recommendations, it will be necessary to implement algorithms that help to deliver good starting visualizations and allow to transfer recommendations between similar data sets. Vartak et al. [26] present an approach to mine statistically significant visualizations from a database. A similar approach could be

adapted to create visualizations for scenarios, where there is a lack of previous interaction data. Chirigati et al. [6] contribute an approach to identify similar data characteristics within the same data set.

Finally, we look at the combination of user interaction and machine learning and at approaches that use machine learning for recommendation systems. In machine learning approaches user interactions are mainly used to integrate human feedback into the learning process and to improve models. In their VA system, Sacha et al. [24] support the user in carrying out suitable machine learning steps by using the semantic model of the data and selecting suitable methods for the corresponding use case. Therefore, they support the user in making decisions through visual guidance and interaction. Brown et al. [3] present an approach for active learning for classification and clustering models. The algorithm queries the user iteratively for labels of unlabeled data points where data points can be selected that provide the greatest possible gain of knowledge. Based on the user selection the automatic algorithm will refine the model after every interaction. Convington et al. [8] show how deep neural networks are used for recommendations on the YouTube video platform. From the available videos, recommendation candidates are generated by using users' viewing history and context. Those videos are then ranked by a machine learning algorithm and presented to the user. From the related work in the area of machine learning, we saw approaches on how to integrate user interaction into machine learning and how to use machine learning for recommendation systems.

## 3 USING MACHINE LEARNING IN ANALYTIC PROVENANCE FOR GUIDED EXPLORATORY ANALYSIS

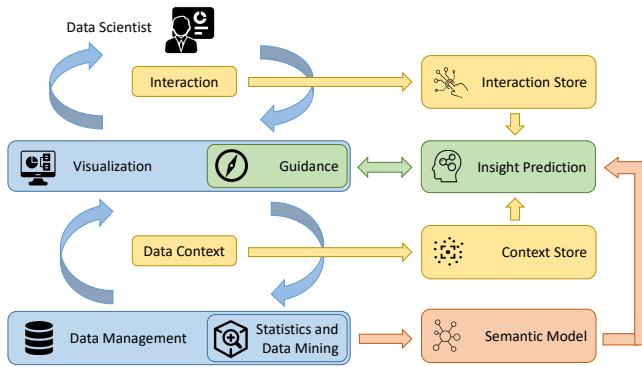
In this section, we describe a concept to implement a machine learning based guidance system that recommends analysis operations based on recorded user interactions and analysis context. We will describe the idea, the system key components and show how they interact with each other.

### 3.1 Idea

There are two core elements that constitute a gain in knowledge in VA. One is an identified, interesting data point and the other is the sequence of operations an analyst has performed to understand the data point and its context [25]. Take, for example, a process from the manufacturing industry in which two parts are first glued and then welded together. Now it can happen that the glue has been applied too thickly and during the subsequent welding process, the seam becomes soiled because the welding head is blocked by the excessive glue. In this case, it is not possible for a data analyst to understand the data of the conspicuous weld seam without also looking at the corresponding data of the gluing process. Therefore, a specific sequence of analysis operations (e.g. creating certain views) is necessary to generate knowledge about conspicuous features in the production process. However, the recommendation must not be made on the statistical selection of analysis methods without considering the context, since processes are different, and the findings may not be transferable to other processes. Both core elements can be learned from domain experts. Domain experts know what

interesting data points look like in their specific domain and they know what sequence of visualizations they need to look at in order to understand them. However, they can also be supported by automatic approaches, e.g. methods to detect interesting data points like outliers. When both elements are combined in an approach for guidance in EDA, there is great potential to maximize the insight gain and accelerate the time to reach it. To implement such a guidance system two key components are necessary that can be derived from similar recommender systems: the analysis operations sequence candidate generation and the insight ranking of the candidates. The candidates with the highest ranking can then be used for guidance in the visualization of an analytics tool.

### 3.2 System Components



**Figure 1: Overview of system components for using machine learning with analytic provenance data for guided EDA.**

Figure 1 gives an overview of all system components that need to be implemented for a first evaluation of the concept. Furthermore, it also shows the interdependencies between the components. A data scientist interacts with the visual representations of the data in the visualization layer.

*Visualizations* can be extended with the addition of new use cases and domains depending on which visualizations are needed by domain experts. The visualization allows to create charts by adding features to axes and visualizing the data by several types of charts. Since the guidance system presented here does not replace statistical and data mining methods to process or aggregate data, those methods must also be available in the system in an extension of the data management layer where data can be selected. The analyst can transfer views into a final report to indicate that insight was generated by the view.

The interactions carried out during the data analysis process are stored in the *interaction store* as an analysis interaction history. The data is stored in graph-based representation to cover all analytic sequences. The interaction store should therefore be seen much less as a collection of individual interactions but more as a semantic sequence of analysis steps, the aim of which is to create deeper insights into the data set. Table 1 shows an overview of an initial interaction set for the application based on interactions proposed by Gotz et al. [12] and Camisetty et al. [5]. The set covers general

analysis interactions of typical VA tool. The interaction store persists the interaction graph in a graph database. That allows for fast querying of specific interaction sequences (e.g. to evaluate occurrences of the specific sequence) and performant manipulation of the interaction graph. The whole sequences can only be evaluated together with its semantic and data characteristics.

| Action type | Interaction              | Parameters            |
|-------------|--------------------------|-----------------------|
| Meta        | Initial                  |                       |
|             | Undo                     |                       |
|             | Redo                     |                       |
|             | Reset                    |                       |
| Analysis    | Add feature to axis      | Feature, axis         |
|             | Remove feature from axis | Feature, axis         |
|             | Chart visualization      | Chart type            |
|             | Apply filter             | Feature, filter range |
|             | Pan                      | Pan range             |
|             | Zoom                     | Zoom range            |
|             | Details                  | Data point            |
| Insight     | Add view to final report |                       |

**Table 1: Initial set of interactions to be tracked by the interaction store.**

The *context store* evaluates the characteristics of the data that are contained in views, which are put into final reports. Those views yield some kind of insight, which is often generated by data with interesting characteristics as spikes or inconsistencies. The context store identifies those data points and searches for similar data points within new data sets. The representation of the characteristic depends on the data type. An approach to identify similar characteristics within data sets has been purposed by Chirigati et al. [6].

In addition to the analyzed data set itself, there will also be a *semantic model* created of the data and the associated domain, which provides further information on the origin and use of the data. Unless otherwise possible, the semantic model must be created by the data scientist when selecting the data set. Management of semantic models can be implemented by a semantic data system as purposed by Pomp et al. [21]. Similar domains will be identified by an overlap in their semantic description.

All three, the interaction store, the context store and the semantic model, serve together as input data for the *insight prediction*. The insight prediction implements a neural network based recommender system. The neural network will be trained on all interaction sequences that end in a state that is put into a final report to indicate insight. For new analyses, the insight prediction performs three steps: In the first step, it will identify interaction data of similar data analyses based on the similarity of the semantic models that are stored in the context store. Afterwards, analysis interaction sequence candidates will be derived from the data characteristics and historical interaction data from the interaction store. In the final step, the candidates will be ranked by predicted insight from the neural network. Based on the model, a guidance system highlights certain analysis steps as part of the visual processing, depending on how much insight would be generated by their execution.

## 4 DISCUSSION

In the following, we want to discuss the presented concept by comparing it to current research, highlight key challenges for a successful implementation and show the path forward.

Current research of analytic provenance focuses on generating optimized guidance systems based on insights gained from the analysis of recorded user interactions. Since domain knowledge is sometimes very complex, this approach could prove to be extremely cumbersome. With recent advances in machine learning methods a technology has become available, that allows us to preserve and harness complex knowledge without understanding its dependencies in detail. On the contrary, this provides a new opportunity to explore conserved knowledge by retrospectively examining the trained models to understand the ways in which people approach EDA. The recommendation of a analysis sequences also offers advantages if a data scientist has already found insights by viewing different ways that other analysts have used to find those insights. That will help to better understand what makes a specific data point interesting in its context. There will also be a need to implement and compare several different methods in this approach. The difficulty here will be to generate a good complete sequence of interaction steps, which then lead to deeper insight into the data set. More advanced methods will be needed to label interaction sequences by the amount of insight they generate to enable better candidate generation and insight prediction. There are some related areas, such as the time series prediction, that also deal with the prediction of several consecutive data points based on historical data. It would also be possible to include direct user feedback in the approach in order to create better models through interactive learning.

A large number of different analytic provenance data sets already exist [16]. However, this data is widely scattered across domains, tools and use cases. This in turn leads us to another challenge, because the amount of data per use case or even domain — i.e. the number of exploratory analyses by experts — will probably be very high to receive decent results and thus difficult to generate for a single use case. The key to this could be the transfer of recommendations between similar domains and use cases. As an approach to enable this, we included semantic models to determine the similarity between use cases. Semantic models have gained in importance, especially in the area of data integration and retrieval, in order to cope with large amounts of heterogeneous data [21]. It could therefore be possible that in the future semantic models of data will be created during the integration.

Another important factor will be the quality of the training data, as it will massively influence the accuracy and thus the usefulness of the predicted actions that will guide analysts to interesting insights. Therefore, it is important to consistently capture and evaluate interactions that are performed during an EDA. Some relevant work exists in the area of semantic user interaction [9, 11]. Interactions should not be interpreted independently, but as a sequence of operations aimed at generating the highest possible amount of insight. However, this is not a purely technical question, since the amount of insight depends strongly on the experience and context of the persons involved. For example, it makes a huge difference what a domain novice needs to learn in a data analysis in contrast to a domain expert. Thus, the way a novice must perform an EDA to

understand insights will be very different from a domain expert. A simple example of this is that the insight an analyst generates through an analysis operation sequence decreases with each time he repeats that analysis. Therefore, it might be interesting to personalize recommendations and to integrate the personal context into the system. As a starting point with each data analysis that generates training data, a qualitative evaluation of the associated EDA should be carried out, for example by interviewing the expert using a questionnaire. This questionnaire must ensure that the expert has in-depth domain knowledge and assesses the analysis. From the self-assessment, we can deduce, for example, whether there was any deeper knowledge within the data and how difficult it was to derive it.

## 5 CONCLUSION AND FUTURE WORK

The recent advance in machine learning technology offers additional opportunities for visual provenance research. While most of the research currently focuses on recording and analyzing user interaction, we presented a concept that applies machine learning to recorded interactions to guide analysts during the sensemaking process. Recorded data include interaction histories of expert users as well as the analysis' context. From the recorded data analysis operation sequence candidates are derived that are then ranked by their predicted insight. We contributed a system design and showed interdependencies between the presented components. Based on this concept, we then discussed core research questions and challenges for further research as well as improvements that this concept can contribute for knowledge transfer between analysts and domain experts. While the investigation of the learned machine learning models will lead to new research opportunities afterwards, the transfer between domains to achieve the required amount of training data and the constant quality assurance of the input data will be decisive factors for the successful implementation itself.

To continue our work, we will implement a proof of concept, collect interaction data for a more complex domain, and evaluate the predictions of the system together with domain experts. After successful evaluation, the interaction data set can then be extended. However, a quality assurance system must first be developed and implemented in order to maintain the quality of the predictions at a high level in the long term. Furthermore, for the implementation, the visual guidance system must also be developed, which adapts the user interface in order to be able to easily carry out further studies in different domains and use cases. A key requirement is to guide the analyst through a series of analysis steps that only executed in consecutive order will yield a specific understanding and insight of the data set. Finally, the most difficult point will probably be to investigate the transferability between domains and use cases. It will also be shown to what extent this is possible using semantic models. In general, after each use case we will check how the trained model is structured and how it evolves with the integration of further domains and use cases.

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